

SIMPLE & MULTIPLE LINEAR REGRESSION

AI & ML Internship – Task 8 Report

1. Introduction

Linear Regression is one of the most widely used supervised machine learning algorithms. It is used to predict continuous numerical values by establishing a relationship between input variables and an output variable. In this task, both Simple Linear Regression and Multiple Linear Regression were implemented using the Advertising Dataset to predict Sales based on advertising expenditure.

2. Objective

- To understand Linear Regression concepts.
- To implement Simple Linear Regression using a single feature.
- To implement Multiple Linear Regression using multiple features.
- To perform Train-Test Split.
- To predict Sales values using regression models.
- To evaluate model performance using Mean Squared Error (MSE).
- To evaluate model performance using R^2 Score.
- To interpret regression coefficients.

3. Dataset Information

Dataset Used: Advertising Dataset
File Format: CSV
Input Features: TV, Radio, Newspaper
Target Variable: Sales

4. Technologies Used

Python
Pandas
NumPy
Matplotlib
Seaborn
Scikit-learn
Jupyter Notebook / VS Code

5. Steps Performed

Step 1: Imported Required Libraries
Step 2: Loaded Advertising Dataset
Step 3: Checked Dataset Information
Step 4: Generated Statistical Summary
Step 5: Checked Missing Values
Step 6: Generated Correlation Heatmap
Step 7: Implemented Simple Linear Regression
Step 8: Applied Train-Test Split
Step 9: Trained Simple Linear Regression Model
Step 10: Predicted Test Data

Step 11: Evaluated Model using MSE and R^2 Score
Step 12: Visualized Regression Line
Step 13: Implemented Multiple Linear Regression
Step 14: Trained Multiple Linear Regression Model
Step 15: Predicted Sales Values
Step 16: Evaluated Model Performance
Step 17: Analyzed Regression Coefficients

6. Output

- Simple Linear Regression model trained successfully.
- Multiple Linear Regression model trained successfully.
- Sales values predicted successfully.
- Correlation analysis performed successfully.
- Regression line visualized successfully.
- MSE calculated successfully.
- R^2 Score calculated successfully.
- Feature importance identified using coefficients.

7. Results

Mean Squared Error (MSE): 3.174
 R^2 Score: 0.8994

Feature Coefficients:

- TV = 0.0447
- Radio = 0.1892
- Newspaper = 0.0028

Observation:

Radio advertising had the highest influence on Sales prediction, while Newspaper advertising had the least influence.

8. Conclusion

In this task, both Simple Linear Regression and Multiple Linear Regression models were successfully implemented using the Advertising Dataset. The relationship between advertising expenditure and Sales was analyzed through regression techniques. The Multiple Linear Regression model achieved an R^2 Score of approximately 0.8994, indicating strong predictive performance. The evaluation results showed that the model was able to explain nearly 90% of the variation in Sales. Coefficient analysis revealed that Radio advertising contributed the most towards Sales prediction, followed by TV advertising, while Newspaper advertising had minimal impact. Overall, this task provided practical understanding of regression modeling, prediction, model evaluation, and interpretation of machine learning results.